

Processes and Threads

A process is a program in execution with its own address space, registers and open files. A thread is the smallest unit of execution that the scheduler can run inside a process. Threads of the same process share memory and open files, which makes communication cheap but requires synchronization. A context switch saves the state of the running process and restores the state of the next one.

CPU Scheduling

The scheduler decides which ready process runs next on the CPU. Round robin gives each process a fixed time quantum and moves it to the back of the ready queue when the quantum expires. Shortest job first minimizes average waiting time but can starve long jobs. Starvation is a situation where a process waits indefinitely because other processes are always preferred.

Virtual Memory and Paging

Virtual memory is a technique that gives each process the illusion of a large, private address space. Paging divides memory into fixed-size pages and frames, and the page table maps virtual pages to physical frames. A page fault occurs when a process accesses a page that is not in physical memory, so the operating system loads it from disk. Thrashing happens when the system spends more time handling page faults than running processes.

Deadlocks

A deadlock is a state where a set of processes are each waiting for a resource held by another process in the set. Four conditions must hold for deadlock: mutual exclusion, hold and wait, no preemption and circular wait. Deadlock prevention breaks at least one of these conditions, for example by ordering resource requests. The banker's algorithm avoids deadlock by only granting requests that keep the system in a safe state.